

Cybersecurity Homelab: Honeypot and SIEM Integration

Project Overview

Designed and deployed a virtualized cybersecurity monitoring environment using Proxmox VE to simulate real-world attack detection and security operations workflows. The lab integrates a Cowrie SSH honeypot with the Wazuh Security Information and Event Management (SIEM) platform to capture, analyze, and investigate attacker activity.

The project demonstrates hands-on experience with virtualization, Linux administration, network services, log collection, security monitoring, and incident investigation.

Objectives

The primary goals of this project were:

- Build a virtualized cybersecurity lab environment.
 - Deploy a realistic SSH honeypot to attract and record attacker behavior.
 - Implement centralized log collection and security monitoring.
 - Forward honeypot events into a SIEM platform.
 - Simulate attacks from an attacker workstation.
 - Gain experience with enterprise security tools and workflows.
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Environment

Hypervisor

- Proxmox VE 9.1.9

Host Hardware

- HP EliteDesk 800 G6 Desktop Mini
- Intel Core i5-10600T (6 Cores / 12 Threads)

- 16 GB DDR4 RAM
- 512 GB NVMe SSD

Virtual Machines

VM	Purpose
Kali Linux	Attacker workstation
Ubuntu Server 24.04.4	Cowrie Honeypot
Ubuntu Server 24.04.4	Wazuh SIEM Platform

Phase 1: Cowrie Honeypot Deployment

Installation

A dedicated Ubuntu Server 24.04.4 virtual machine was created to host the Cowrie SSH honeypot.

Configuration

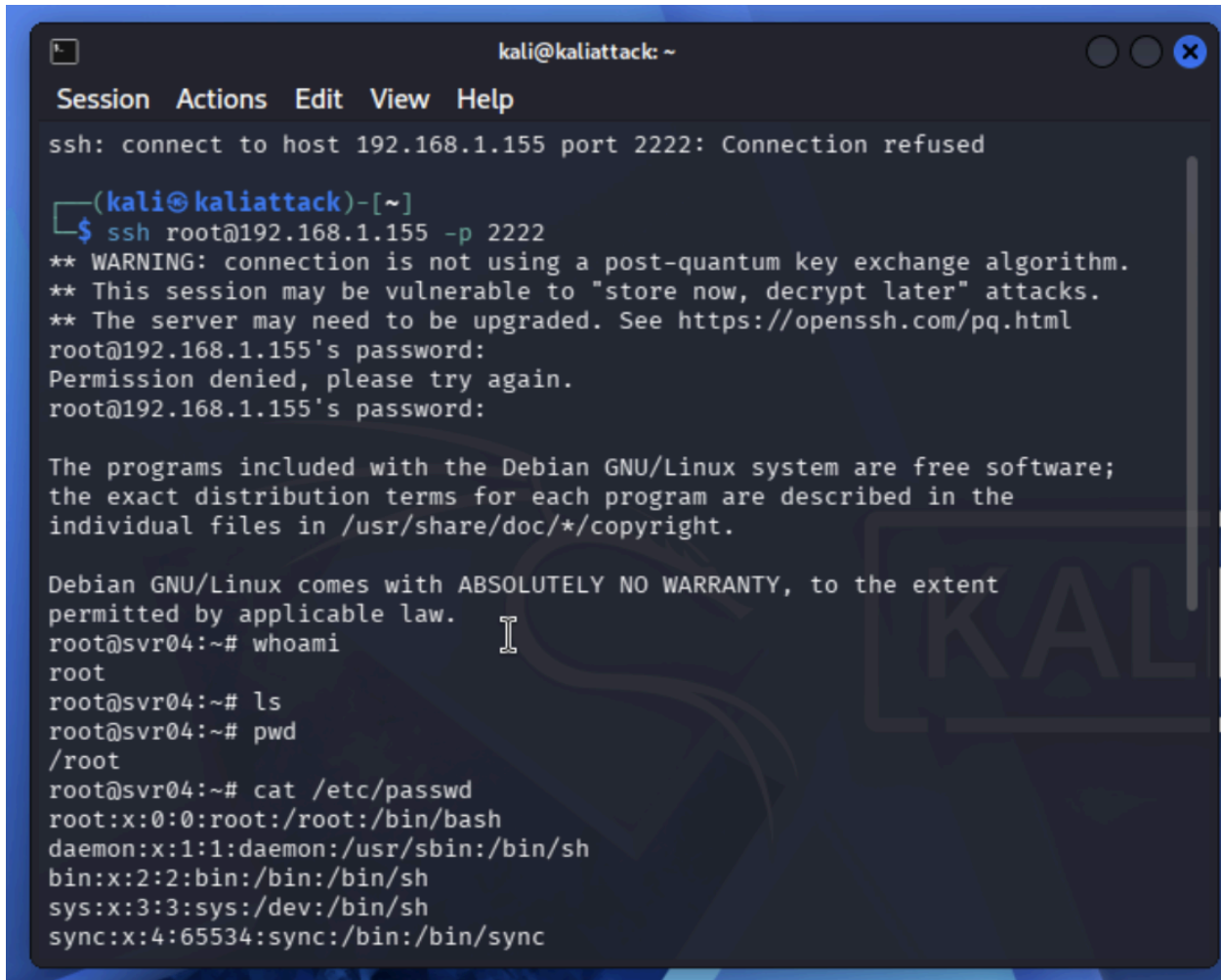
- OpenSSH installed
- Python virtual environment created
- Cowrie cloned from GitHub repository
- Dependencies installed using pip
- Dedicated Cowrie user created
- Cowrie configuration generated and customized

Validation

Verified successful operation by:

- Launching Cowrie services
 - Monitoring Cowrie log files
 - Testing SSH connections from Kali Linux
 - Confirming attacker activity was recorded
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Attack Simulation



```
kali@kaliattack: ~
Session Actions Edit View Help
ssh: connect to host 192.168.1.155 port 2222: Connection refused

(kali@kaliattack)-[~]
$ ssh root@192.168.1.155 -p 2222
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
root@192.168.1.155's password:
Permission denied, please try again.
root@192.168.1.155's password:

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@svr04:~# whoami
root
root@svr04:~# ls
root@svr04:~# pwd
/root
root@svr04:~# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
```

Using the Kali Linux virtual machine, multiple SSH login attempts were performed against the honeypot.

Example Activities

- Failed root login attempts
- Successful login using Cowrie's fake credentials
- Command execution within the fake shell

Commands tested included:

```
whoami
ls
pwd
cat /etc/passwd
```

Cowrie successfully captured:

- Source IP addresses
 - Login attempts
 - Usernames
 - Passwords
 - Commands executed by the attacker
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Phase 2: Wazuh SIEM Deployment

Installation

A dedicated Ubuntu Server 24.04.4 virtual machine was deployed for Wazuh.

Resources

- 6 GB RAM
- 80 GB Storage
- 2 vCPUs

Components Installed

- Wazuh Manager
 - Wazuh Indexer
 - Wazuh Dashboard
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Storage Expansion

During deployment, additional storage requirements were identified.

Tasks completed:

- Expanded VM disk size to 80 GB
- Resized Logical Volume Manager (LVM) partitions
- Extended filesystem capacity
- Verified available storage using:

`df -h`

pvs
vgs

This increased available storage from approximately 38 GB to 76 GB.

Phase 3: Agent Integration

Agent Deployment

Installed the Wazuh Agent on the Cowrie server.

Challenges Encountered

During deployment, several issues were identified and resolved:

Version Mismatch

Issue:

Agent version must be lower or equal to manager version

Resolution:

- Removed Wazuh Agent 4.14.5
- Installed matching Wazuh Agent 4.13.1

Configuration Error

Issue:

Invalid server address found: MANAGER_IP

Resolution:

- Updated agent configuration
- Replaced placeholder values with actual Wazuh Manager IP address

Duplicate Agent Registration

Issue:

Duplicate agent name

Resolution:

- Verified existing registration
 - Confirmed successful communication between manager and agent
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Phase 4: Log Integration

Cowrie to Wazuh Integration

Configured the Wazuh agent to monitor Cowrie's JSON log output.

Added custom log monitoring configuration:

```
<localfile>  
  <log_format>json</log_format>  
  <location>/home/cowrie/cowrie/var/log/cowrie/cowrie.json</location>  
</localfile>
```

This enabled Wazuh to ingest and process Cowrie security events.

Results

Successfully achieved end-to-end visibility of attacker activity.

Security Workflow

Kali Linux



Cowrie Honeypot
↓
Cowrie JSON Logs
↓
Wazuh Agent
↓
Wazuh Manager
↓
Wazuh Dashboard

Detection Capabilities Demonstrated

The environment successfully recorded:

- SSH brute-force attempts
- Failed authentications
- Successful honeypot logins
- Command execution activity
- Source IP attribution
- Session logging

All events were visible through the centralized Wazuh dashboard.

Skills Demonstrated

Cybersecurity

- SIEM Administration
- Security Monitoring
- Honeypot Deployment
- Log Analysis
- Threat Detection
- Incident Investigation

Networking

- Network Bridging
- Virtual Networking

- SSH Services
- IP Configuration

Linux Administration

- Ubuntu Server
- Service Management
- User Management
- Package Management
- Log Monitoring

Virtualization

- Proxmox VE
 - VM Provisioning
 - Resource Allocation
 - Snapshot Management
 - Storage Expansion
-

Future Enhancements

Planned improvements include:

- pfSense firewall deployment
- Active Directory integration
- Windows endpoint monitoring
- Network segmentation
- Internet-facing honeypot deployment
- Multi-agent Wazuh monitoring
- SOC dashboard customization
- Threat intelligence integration